NAME: ABDULLAH **ROLL NO: 044**

CLASS: BSAI **SECTION: 3A**

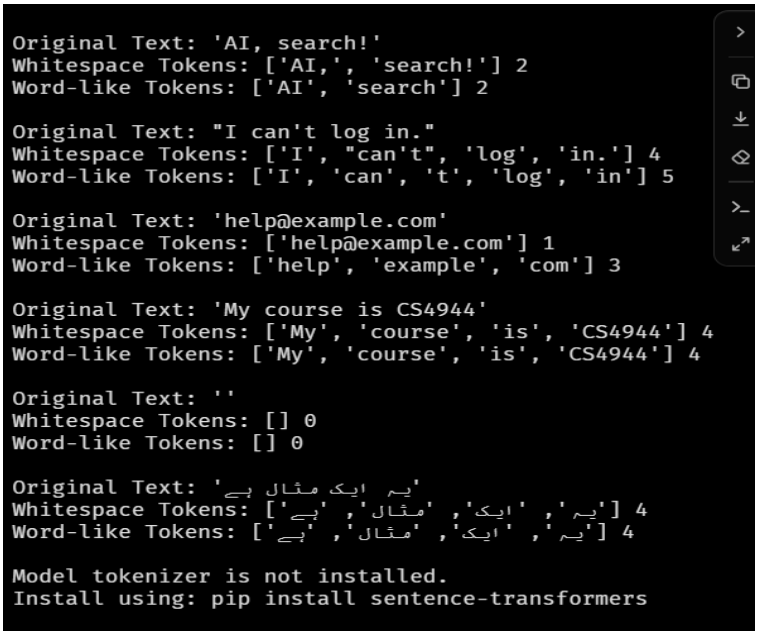
LAB TASK NO: 6

QUESTION:1)

```
import re
samples = [
    "AI, search!",
    "I can't log in.",
    "help@example.com",
    "My course is CS4944",
    "",
    "یہ ایک مثال ہے"
]
for text in samples:
    ws = text.split()
    words = re.findall(r"\b\w+\b", text, flags=re.UNICODE)
    print("\nOriginal Text:", repr(text))
    print("Whitespace Tokens:", ws, len(ws))
    print("Word-like Tokens:", words, len(words))
assert re.findall(r"\b\w+\b", "") == []
try:
    from sentence_transformers import SentenceTransformer
model = SentenceTransformer(
        "sentence-transformers/all-MiniLM-L6-v2",
        device="cpu”)

    print("\nModel Tokenizer Results")
 for text in samples[:2]:
```

```
    print(text, model.tokenizer.tokenize(text))
except ImportError:
    print("\nModel tokenizer is not installed.")
    print("Install using: pip install sentence-transformers")
```

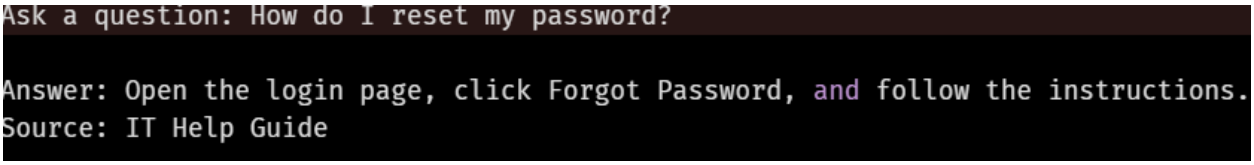


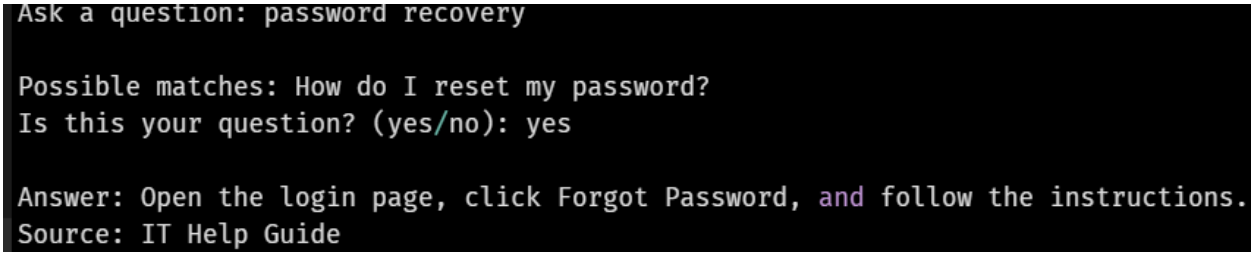
FAQ Chatbot

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
faqs = [
    {"question": "How do I reset my password?",
        "answer": "Open the login page, click Forgot Password, and follow the instructions.",
        "source": "IT Help Guide”},
    {
        "question": "How do I install Python?",
        "answer": "Download Python from its official website and follow the installation steps.",
        "source": "Python Installation Guide"
    }]
```

```
questions = [faq["question"] for faq in faqs]
vectorizer = TfidfVectorizer()
faq_vectors = vectorizer.fit_transform(questions)
def reply(user_input):
    user_input = user_input.strip()
 if user_input == "":
        return "Please enter a question."
for faq in faqs:
        if user_input.lower() == faq["question"].lower():
            return "Answer: " + faq["answer"] + "\nSource: " + faq["source"]
user_vector = vectorizer.transform([user_input])
    scores = cosine_similarity(user_vector, faq_vectors)[0]
best_index = scores.argmax()
if scores[best_index] >= 0.15:
        return "Possible matches: " + questions[best_index]
return "No matching FAQ found."
if __name__ == "__main__":
    print("Welcome to FAQ Chatbot!")
while True:
        question = input("\nAsk a question: ")
        result = reply(question)
print(result)
 if result.startswith("Possible matches:"):
            confirm = input("Is this your question? (yes/no): ")
If confirm.strip().lower() == "yes":
                matched = result.replace("Possible matches: ", "")
for faq in faqs:
```

```
            if faq["question"] == matched:
                print("Answer:", faq["answer"])
                print("Source:", faq["source"])
                break
else:
        print("Please ask another question.")
again = input("\nAsk another question? (yes/no): ")
 if again.strip().lower() != "yes":
        break
```





Written explanation

Q1) Two differences between whitespace tokenization and regex tokenization are:

Whitespace will split at spaces, while regex can split punctuation and emails.

Whitespace will not split can't while regex will split ‘can’ and ‘t’.

Q2) One issue with normalisation is that the meaning of the text might be changed when removing punctuation or changing the words.

Q3) The chatbot asks for confirmation as it uses TF-IDF to get the best match. If there isn’t a perfect match it will suggest the possible FAQ answer and ask for confirmation so they don’t give the wrong answer.

Q4) No. Regex can identify Unicode such as words but does not take into consideration all language specific tokenizations. **GIT HUB LINK:**

https://github.com/abdullahrehman957-lgtm/AI-LAB-TASK